

**STATISTICAL ANALYSIS OF GPS TEC ANOMALIES
ASSOCIATED WITH EARTHQUAKES AND A CASE STUDY OF
MW>7 EARTHQUAKES IN JAPAN**

by

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**Institute of Space Technology,
Islamabad, Pakistan
2022**

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A thesis submitted to the

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In partial fulfillment of the requirements

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by

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by

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ABSTRACT

The GNSS TEC data provide an insight into Ionosphere anomalies during Seismic Events. To find the Correlation between them, two separate studies were conducted on certain criteria. In the first study, four Earthquakes were occurring of $M_w > 7$ from 20 February to 14 April 2011 in a coincided area of Japan. Where GPS TEC data was obtained from three IGS Stations in the EQs vicinity. The TEC variations were assessed with EQs, their date of occurrence, depth, and magnitude along with geomagnetic activity during the period. The Geomagnetic activity overshadows the Ionosphere Anomalies and their association with EQs. However, TEC value abruptions started on 8 March and continues for 6 days in March during which the three EQs occurred. The other study focused on 11 EQs from 2018, in which TEC data examine the Quiet Geomagnetic activity. Where the TEC data were obtained from at least three IGS stations in the surrounding area of EQs Epicenters in different regions of the Globe. The study shows significant Ionosphere deviation before EQs with an average of 4 TECU and a percentage of 44%. However, the TEC variations were not found to be proportional to the depth of an EQ. Due to the Complex Nature of the Phenomenon, TEC values cannot be defined as a reliable precursor for EQ future forecasting.

Keywords:

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LIST OF ABBREVIATIONS

GNSS	Global Navigation Satellite System
GPS	Global Navigation System
GALILEO	European Global Satellite Navigation System
GLONASS	Global Navigazion naya Sputnikovaya Sistema
IRNSS	Indian regional navigation satellite system
Δ TEC	The difference in total electron content
USA	United States of America
EU	European union
TEC	Total Electron Content
SW	Solar Wind
QZSS	Quasi Zenith Satellite system
IRNSS	Indian regional navigation satellite system
CMEs	Coronal Mass Ejections
EQs	Earthquakes
IIP	Ionospheric Piercing Point
CID	Coseismal Ionospheric Disturbances
Dst	Disturbance storm index
Kp	Planetary K index
AP	The earliest occurring maximum 24-hour value is obtained by computing an 8-point running average.
IGS	International GNSS Service
RINEX	Receiver Independent Exchange Format
USGS	United States Geological Survey
VTEC	Vertical total Electron Content
IQR	Inter Quartile Range
UTC	Universal Coordinated Time
GIM	Global Ionospheric map

1 INTRODUCTION

1.1 Global Navigation Satellite Systems

The Global Navigation Satellite Systems is the branch of Geoinformatics, which primarily provides Geo-Spatial Positioning over the Entire Globe. Over the years several international and regional systems are developed. The Global Positioning System (GPS) is designed by the USA and was completed early, covering the whole of Earth. The civilian signals provided by GPS are free of cost and available to everyone in any country. Other systems include Galileo by the Europe Union, GLONASS by Russian Federation, and Compass (BeiDou) by China. Some Regional Systems include QZSS of Japan and IRNSS of India.

1.2 Structure

The GNSS structure consists of three Segments:

- The space segment consists of a Constellation of Satellites, which revolve around Earth's surface, and transmits the signal in the range of Microwave of at least two frequencies.
- In the control segment, signals are monitored, calculations are made, and navigation data is uploaded to the satellite to maintain the health of the system. It consists of multiple ground stations that receive the signals and perform satellite monitoring, including satellite health, orbital information, and clock corrections to the satellite. This status is forwarded to the command and control stations that correct the ephemeris and clock of the satellite as well as perform maneuvers in case of satellite trajectory deflection.
- The user segment consists of end-users of GNSS technology that receive the signals and extract position and time information for the use of different sectors, such as banking and autonomous vehicles. grid stations, military and defense purposes, etc.

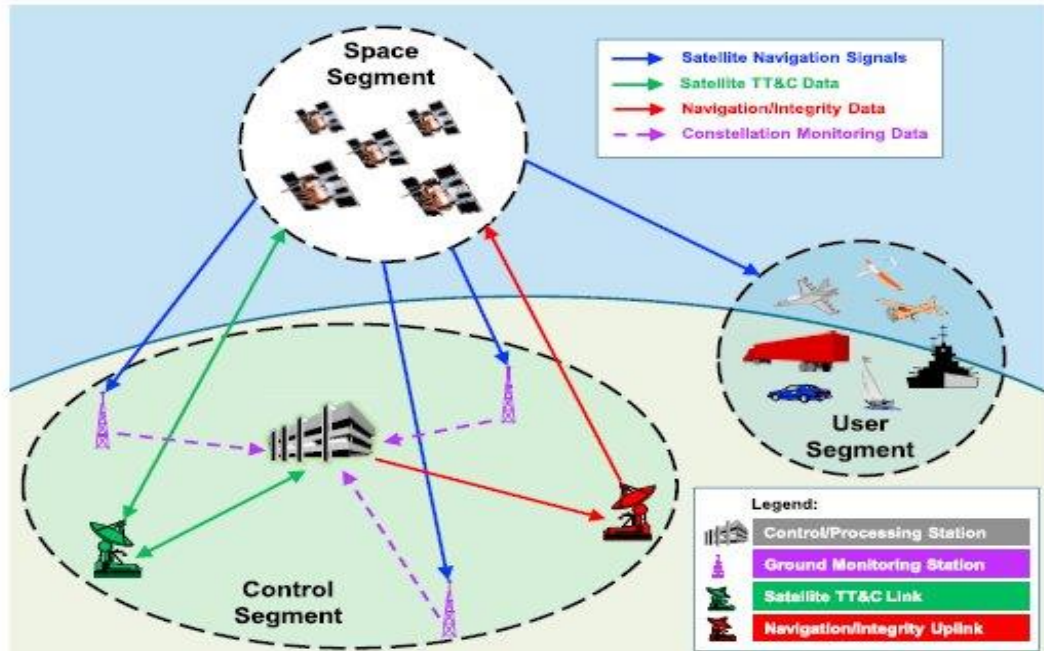


Fig. 1.1. GNSS Structure containing the (a) Space Segment (b) Ground Control Segment (c) User Segment
(Courtesy of Sciencedirect.com)

1.3 Applications

The availability of GNSS signals for navigation and timing purposes produced a wide range of applications for commercial and civilian purposes. The positioning accuracy of a surveyor or mapper is measured in centimeters or sub-centimeters in high-end applications such as land and hydrographic surveying. The accuracy of a low-end application such as personal navigation or another mass-market (consumer) application is a few meters.

In the aviation community, GNSS has been embraced because it offers a high-performance global navigation capability that meets many of the aviation users' needs. Additionally, GNSS receivers are inexpensive, lightweight, and flexible, which makes them an attractive option for navigation.

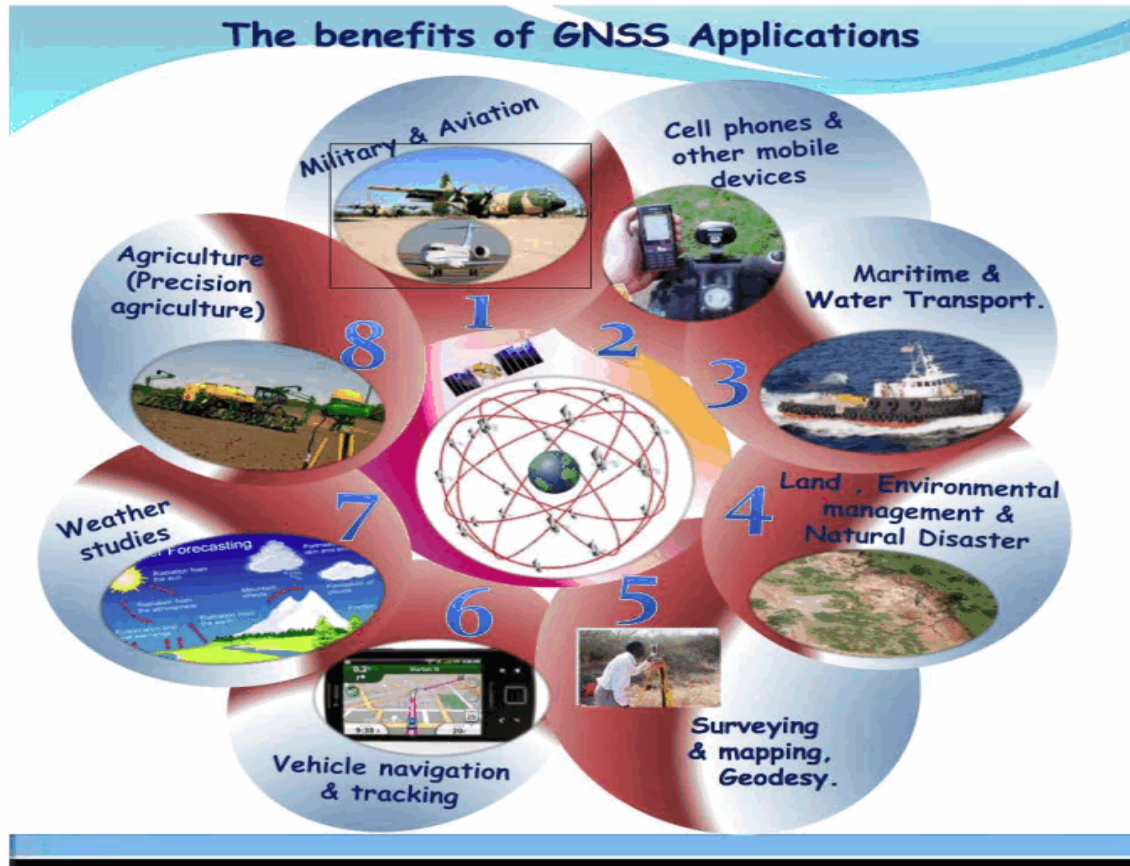


Fig 1.2 GNSS Applications in different fields.

(Ministry of agriculture Botswana)

Geodetic surveyors were early civilian GPS users, who employed technology to identify the positions of ground markings in control networks. Now, GNSS is the primary Technology for Geodetic surveying. In the event of tectonic plate movements, plate boundary deformation, earthquakes and seismology, volcanic deformation, surface loading deformation, and glacial isostatic adjustment, GNSS aids in the study of Earth's geodynamics.

1.4 Ionosphere and TEC

In satellite communication, the signal traveling time is crucial for accurate data extraction. In the case of GNSS, the signal time is important for ranging the distance between the satellite and the receiver. The speed of the signal in lower space is almost equal to the speed of light, which is of any electromagnetic wave speed. However, in the ionosphere, certain

elements delay the signal time to a significant proportion. Ionosphere makeup is important to know for GNSS users.

1.5 Ionosphere

The ionosphere is situated in lower space ranging from 60km to 1000km from the surface of the earth. Some noble and other gases are present in this part of space from the earth's atmosphere. In addition, during daytime or Solar storms, the incoming solar rays polarize the gases present in the Ionosphere. It consists of four-layer named D, E, F1, and F2 at different heights. The D layer ranges from 60 km to 90 km, the E layer exists between 90 km to 150 km, and above 150 km the F layers occur. During the solar activity, the Sun's ultraviolet rays polarized the gases such as nitrogen, and oxygen, at lower heights and helium and hydrogen at upper heights. This causes the gaseous atoms and molecules to form ionized charged particles.

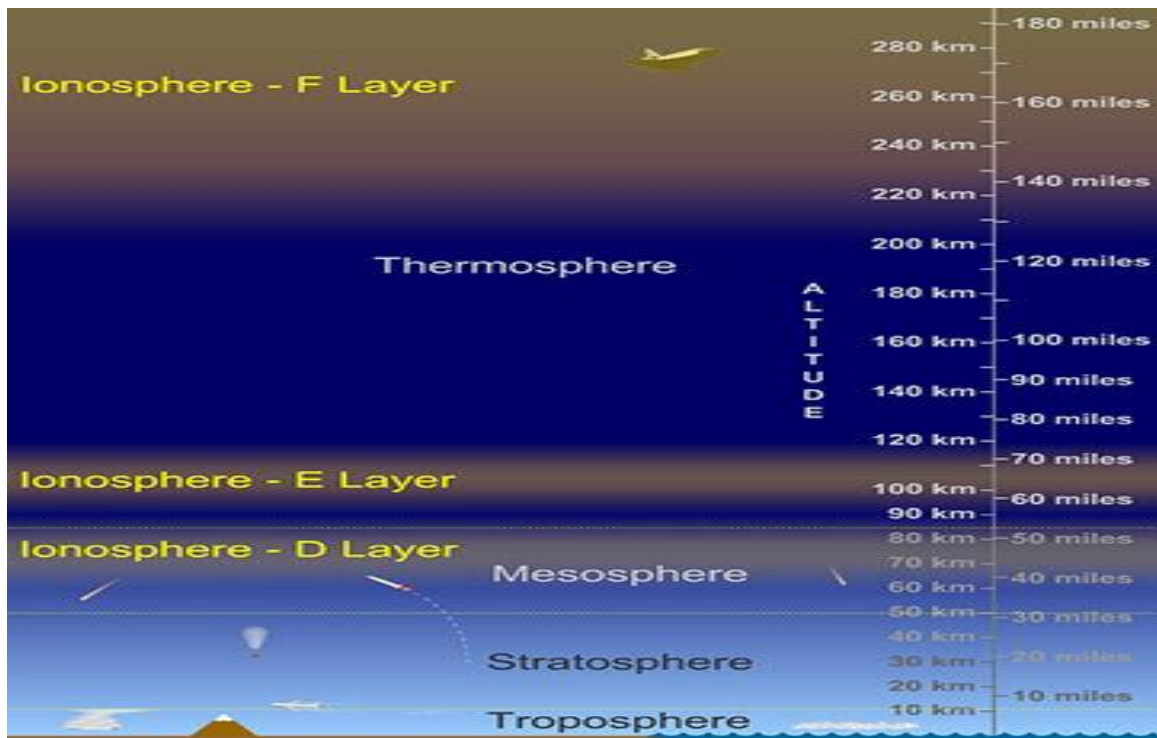


Fig 1.3 Layers of Ionosphere at different altitudes

(Courtesy: Center for Science Foundation)

The other factor is ionosphere and thermosphere coupling, in which the plasma transfers from the F2 layer. This process occurs by diffusion and neutral winds, which are largely dependent on thermosphere conditions. The formation of ions and electrons was caused by

collisions between charged and neutral particles.

1.6 Total Electron Content

Total electron content(TEC) is the ionosphere characteristic that has the greatest impact on GNSS signals. A value of TEC equal to 1×10^{16} electrons/m² is called one TEC unit. Values of TEC from 10^{16} to 10^{19} electrons per m² along the radio wave path. Represent the extreme of the observed values in Earth's Ionosphere. TEC measurements are often conducted using available satellite radio frequencies. Random elevation angles were observed. At a mean ionosphere height, the TEC is often stated as an equivalent vertical TEC divided by a slant TEC by the secant of the elevation angle. often traveled between 350 and 400 kilometers. The slant values seen at different elevation angles can be compared to the vertical electron content measurements.

1.7 Lower Space Weather

The lower space weather is largely affected by solar activity. The sun emits electromagnetic radiations that carry plasma and ions. The solar intense activity includes solar flares and coronal mass ejections. When abrupt energy is released from the sun's surface in huge quantity then it is called solar flares. These solar flares can reach the earth's atmosphere in minutes and ionize the gases in the ionosphere. The magnetic disturbances on the sun cause solar flares and produce sunspots on the sun. sunspots have relatively cool temperatures the rest of the surface of the sun reaching up to 4000k. The sunspots may last for days and months.

In the surrounding of the sunspots, magnetically active regions produce coronal mass ejections, which carry plasma and highly energetic particles. A single burst may include billion tons of plasma, which largely misses the earth's atmosphere, and some end in polar regions in the form of aurora. A significant quantity of very energetic particles from CME collisions with the magnetosphere of the earth is absorbed by the upper atmosphere, causing geomagnetic storms, which are perturbations of the magnetic field of the planet. The geomagnetic storm has several indices, which include Dst, Kp, AP, and AE. The disturbance storm index (Dst) gauges the weakening of the equator's magnetic field's current ring. Sudden reduction in the Dst index shows severity in a geomagnetic storm.

Table (1.1) Phases of Geomagnetic Index

Geomagnetic Index	Quiet	Low	Moderate	Strong	Intense
Ap	≤ 20 nT	≥ 30 nT	≥ 50 nT	≥ 80 nT	≥ 100 nT
Kp	0	≤ 3	4	5-6	7-9
Dst	≥ -20 nT	≤ -30 nT	≤ -50 nT	≤ -100 nT	≤ -200 nT
AE	≤ 200 nT	≤ 400 nT	≤ 800 nT	≥ 1000 nT	≥ 2000 nT

1.8 Effects on Satellite Signals in the Ionosphere

The satellite signals that pass through different layers of the ionosphere have a substantial effect. The effects can be reflection, refraction, and absorption of the radio signals. The D region, which extends from 50 to 90 km, does not affect GNSS signals and vanishes during nighttime. The E region, which is also a daytime phenomenon, lies in a range of 90 to 120 km. Additionally, it doesn't have much of an impact on signals, although it can make them scintillate. The F region, which covers a distance of 120 to 1000 kilometers, is the area most affected. The ionization phenomena in the atmosphere are most intense in the F region. The F region is divided into two layers F1 and F2. The F1 is the lower part of the region and is active during the summer. Both layers fuse at night. Above the F layer, everything is completely ionized. It is referred to as the photosphere or the H area.



Fig. 1.4 Effects of Solar Storm on Satellite Signals

(Courtesy: nasa.org)

The ionosphere is a heterogeneous layer. It has a constant electric flux. Its effects fluctuate with latitude on earth. The polar regions of the earth can experience unusually strong ionosphere disturbances. However, the greatest TEC values and the broadest horizontal gradient areas are found in the range of around 60° geomagnetic latitude. In relation to the magnetic equator of the Earth, that band is located at 30° N and 30° S. The longer the signal in the ionosphere, the greater the Ionosphere effect on it, due to satellite elevation, that is why near the horizon signals are canceled.

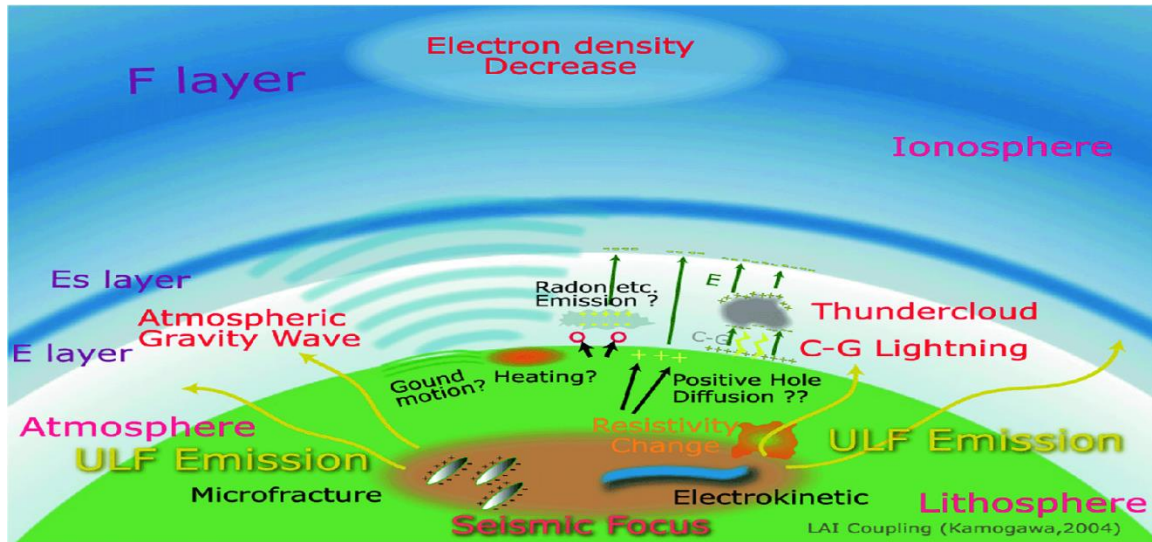


Fig. 1.5 Seismic Activity effect on the ionosphere

(Courtesy: Springer.com)

1.9 Group delay and Phase Delay

Depending on the frequency of the transmission, the ionosphere, a scattered medium, may cause a time delay. Group delay is a term used to describe how long the P-code, C/A code and navigation messages take to transmit. The navigation signal passing through the dense Ionosphere channel experiences a delay due to the change in speed. This causes phase differences, which is known as phase delay. The phase delay increases negatively. In the equation of correction to the signal, the algebraic sign d_{ion} is negative in the carrier phase equation and positive in the pseudo-oranges equation. In other words, the distance between a satellite and a receiver estimated by a code observation will be somewhat too long, whereas the distance established by a carrier observation will be slightly too short.

1.10 Frequency

The effects for frequency vary with magnitude, with more for lower frequency than higher frequency. It indicates that L1, 1575.42 MHz is not impacted as much as L2 1227.60 MHz, and L2 is not affected as much as L5. 1176.45 MHz, indicating that dual-frequency has an advantage over single frequency in this respect. The distances between L1 and L2, L1 and L5, and L2 and L5 are far enough apart to make it easier to estimate ionosphere group

delay. Therefore, a multi-frequency receiver can have produced and remove some ionosphere bias by tracking all the carriers. The following formula explains how the frequency dependency of the ionosphere effects works [28].

$$V = \frac{40.3}{cf^2} TE \quad 1.1$$

Where

V=the Ionosphere delay

c= the speed of light in meters per second

f=the frequency of the signal in Hz

TEC=the quantity of free electrons per square meter

The formula demonstrates that the relationship between time delay and frequency square is inverse. The delay decreases with increasing frequency.

1.11 Coseismal Ionosphere Disturbances

The Coseismal Ionosphere Disturbances (CID) were derived using a global positioning system (GPS). Ionosphere Piercing Point (IIP) height estimates of these disturbances are provided. Because of this, the tectonic forcing may always be immediately linked to the Characteristics of (CID). Compressional waves are created in the surrounding atmosphere by the Earth's crust lifting during an earthquake. In the zone of exponentially diminishing neutral density of the atmosphere, these waves go upward. So, as atmospheric heights grow, does their amplitude. The waves redistribute coseismal ionosphere disturbances, which are ionosphere electronic density perturbations, as they reach atmospheric heights (CID). Significant crustal uplift is caused by thrust earthquakes, whereas strike-slip events mainly cause horizontal deformation of the crust. To explore the CID properties, a variety of ionosphere-sounding methods can be used. The seismic source area receives extensive geographical and temporal coverage from the TEC. The growth of seismic/tectonically induced ionosphere perturbations is heavily influenced by the non-tectonic driving mechanism of satellite geometry, geomagnetic field acoustic wave coupling, and the ionosphere's ambient ionization level. A proprietary acoustic ray-tracing model that was created in-house is used to quantify the impacts of these non-tectonic driving processes at

ionosphere altitudes.

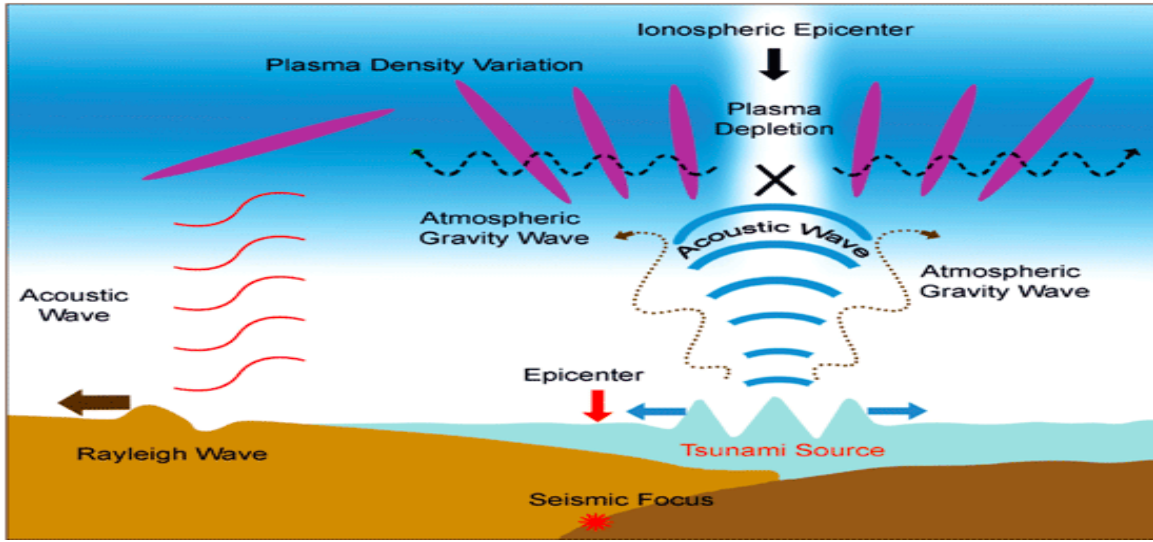


Fig. 1.6 Coseismic Ionosphere Disturbances

(courtesy: Princeton.edu)

1.12 Pre-Seismic Ionosphere Anomalies

Pre-seismic ionosphere anomalies are the anomaly of TEC that occur before and after an earthquake. Many researchers using different approaches have investigated electromagnetic precursors of earthquakes. Pre-seismic TEC anomalies can be divided into two different approaches, the short-term anomaly, and the long-term anomaly. The **short-term anomaly** can be found by using a method used by Ozeki and Heiki [34]. For modeling of background Ionosphere anomaly.

$$\text{Slant TEC } (t, \zeta) = \frac{VTEC(t)}{\cos \zeta + d} \quad (1.2)$$

With the use of this formula, a model of the TEC's raw data is drawn. The bias constant, which is unique to each pair of the satellite station, is given by Time function t , where is the ζ angle of the line of sight and d is the local zenith. By adjusting the least squares, the VTEC is determined. The cubic function of time may be used to describe the VTEC variations during hours.

$$VTEC(t) = at^3 + bt^2 + ct + d \quad (1.3)$$

Using the least-squares approach, variables a, b, c, d, and e may be determined. EQ-related disruptions should be kept to a minimum, some time must be excluded to gain a reference curve. The time 40 mint before and 20 mint after the main shock must be excluded. For the detection of a long-term anomaly, the median of vertical TEC needs to be calculated for the prior 15 days. The next step is fining deviation on the 16th day. If the observed TEC is greater or less than UB and LB in a quiet geomagnetic period, then it is a significant anomaly.

$$LB = M - 1.5 (M - LQ) \quad (1.4)$$

$$UB = M + 1.5 (UQ - M) \quad (1.5)$$

Whereas

LQ = Lower Quartiles.

UQ = Upper Quartiles.

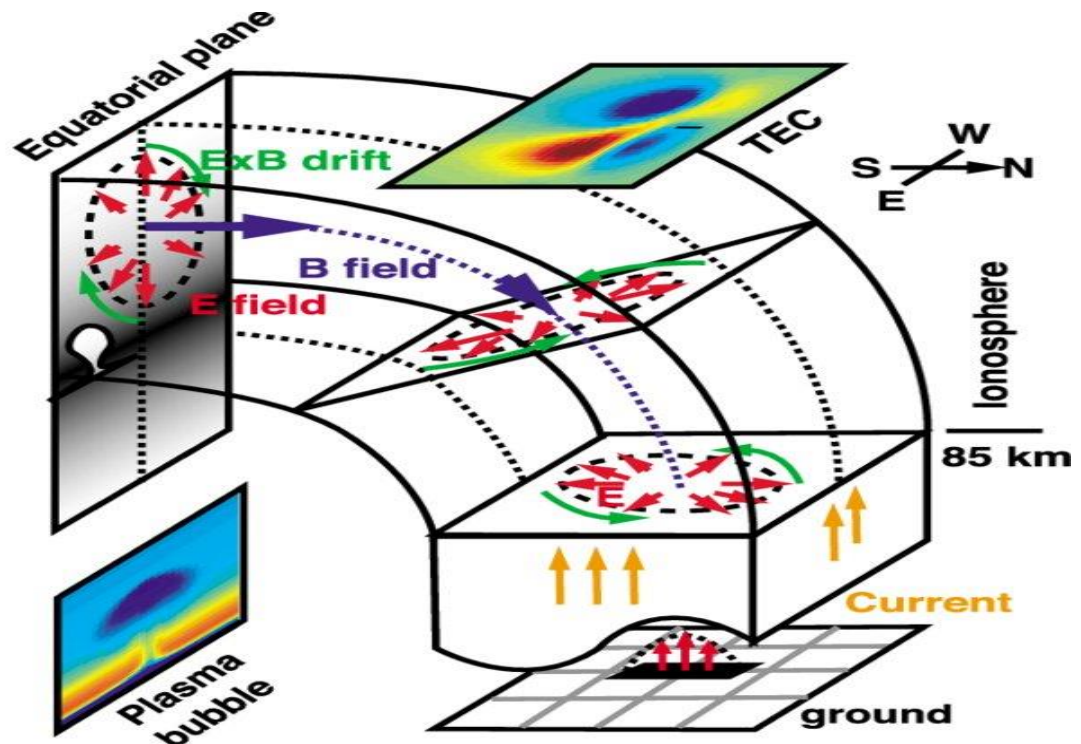


Fig.1.7 An illustration of the electric interaction between the ionosphere and surface charges over earthquake fault lines. The current in the atmosphere (shown by orange arrows) and the electric field E_z at 85 km altitude in the ionosphere are both driven by a vertical surface electric field (shown by red arrows) produced from surface current density J rock across a fault zone. Electric fields cause the $E \times B$ motion (green arrows) of ionosphere plasma, resulting in plasma vortex formation and TEC variations. As a result of the external electric field, a motion eastward occurs in the equatorial plane, leading to the formation of nocturnal plasma bubbles.

(Courtesy: Civildaily.com)

2 Data and Methodology

2.1 Case study-1

The thesis investigates the Ionosphere conditions of four low-intensity earthquakes of $M_w > 7$ before and before the main big earthquake of Tohoku 9.1 magnitude March 11, 2011. The United States Geological survey provided information on EQs. There were few low magnitude shocks on these consecutive days. The selection is based on the condition of magnitude and Time. The magnitude and depth are considered to find the possible relation between earthquake occurrence time with TEC changes in the ionosphere. Three stations were taken near the epic center for TEC data. The earthquake preparation zone is made by the Dobrovolsky formula.

$$R = 10^{0.43M} \text{ km} \quad (2.1)$$

M = Magnitude of EQ.

R = Radius of intense region.

Table (2.1). EQs in Japan region during March- April 2011.

Event	Region	Mw	Date	Time	Location	depth
1	120 km southeast of funato, japan	7.3	09/03/2011	02:45:20 UTC	38.435°N 142.842°E	32.0 km
2	47 km east of arai, japan	7.9	11/03/2011	06:15:40 UTC	36.281°N 141.111°E	42.6 km
3	272 km ES of Kamaishi, japan	7.7	11/03/2011	06:25:50 UTC	38.058°N 144.590°E	18.6 km
4	29 km ES of Ishinomaki, japan	7.1	07/04/2011	14:32:43 UTC	38.276°N 141.588°E	42.0 km

2.2 Geomagnetic Indices

The geomagnetic indices have an effective role in the determination of seismic events. It describes whether the ionization activity in the ionosphere occurs by a solar storm or seismic activity. To carry the research further on seismic events, geomagnetic indices will tell how quiet the solar activity is during the period. Therefore, the earth-bound ionosphere disturbances can be studied. The Kp index and Dst index data were considered during the period of given EQs. The International Service of Geomagnetic Indices provided the data for the geomagnetic indices. Several C, M, and X class flares have been analyzed.

2.3 GPS TEC Data

The TEC data is drawn from GPS satellites, which orbit at an altitude of 20,000 km. these satellites transmit signals via two L bands (1.2 GHz and 1.5GHz) and it passes through the ionosphere to the ground receivers. For robust accuracy, the ionosphere delays are mitigated by the generation of an Ionosphere free combination of two-carrier phases in linear form.

$$L3 = f_{12} / (f_{12} - f_{22}) L1 - f_{22} / (f_{12} - f_{22}) L2$$

The f_1 is the frequency of L1 and f_2 is the frequency of L2 carrier waves. For the study of the Ionosphere, the Phase differences of both frequencies are combined in linear form. These electromagnetic waves face delay in the ionosphere. The delay depends on the size of the frequency, lower frequencies get more ionization and delays. The Temporal changes of TEC along the line of sight(LOS) also called Slant TEC, can be monitored by tracking the difference in L1 and L2 phases. The TEC is measured in TEC units and 1 TEC is equal to $10^{16} \frac{el}{m^2}$. It can be defined as “The total number of electrons integrated between two points, along a tube with $1m^2$ cross-section. After processing, e obtains absolute TEC on removing ambiguities such as cycle slip and inter-frequency biases in-phase data. The TEC is obtained from RINEX (Receiver independent exchange format) files of GPS. It is the exchange format of GPS observation data, which is usually used for post-processing. The coordinate sin RINEX is transformed into an earth-fixed system for the estimation of GPS satellite's orbits. These coordinates are used for the calculation of the SIP (Sub Ionosphere point) trajectory. The following figures show the general concept of SIP and IPP.

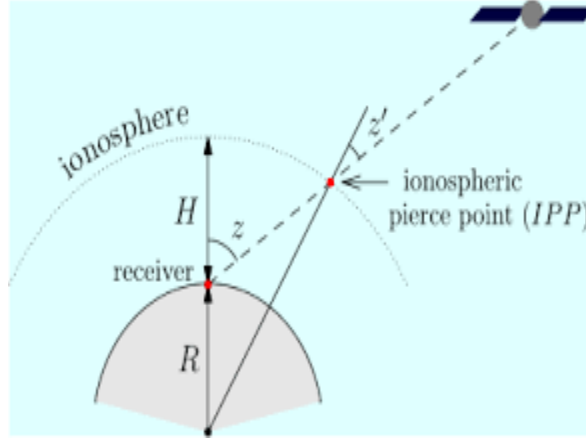


Fig. 2.1 Ionosphere pierce point

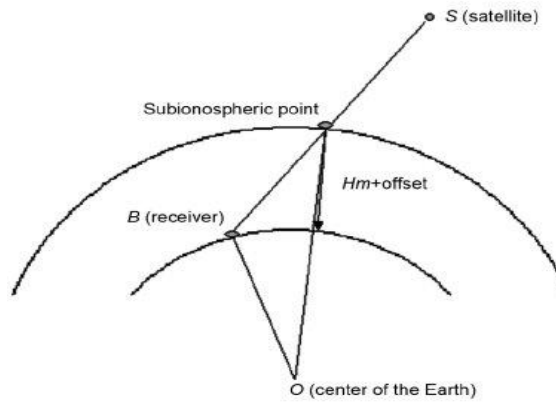


Fig. 2.2 Sub Ionosphere point

(Courtesy: wikipedia.org)

To obtain L_4 , L_1 and L_2 convert to length from radians by multiplying the wavelength of each carrier. L_4 is the difference between L_1 and L_2 . The TEC is gained from L_4 from multiplication with a certain factor.

$$L_4 = L_1 - L_2 \quad (2.3)$$

$$TEC = (1/40.308) f_{12}^2 f_{22} / (f_{12} - f_{12}) L_4 \quad (2.4)$$

Three IGS Stations were chosen for TEC data, the Total Electron content tells us the Ionosphere conditions in these four EQs. Given stations were AIRA, KSMV, and MTKA with a resolution of 2:30 minutes from date 20/02/2011 to 14/04/2011.

2.4 Statistical analysis for Anomalies Detection

To find the anomalies in these EQs, TEC data is statically analyzed. The TEC data is obtained from three IGS Stations, which are dual-frequency receiver stations in the earthquake preparation zone. This TEC data have 2:30 minutes resolution. For static analysis, the confidence bound method has been applied. These bounds are calculated from the median and interquartile range. Below the equation are the mathematical form.

$$UB = (X) + (1.5 \times IQR) \quad (2.5)$$

$$LB = (X) - (1.5 \times IQR) \quad (2.6)$$

Equation (2.5) determines the upper bound and equation (2.6) determines the lower bound on the normal distribution of TEC. Where the exceeding of TEC values from these bounds accounts for abnormality in EQs.

2.5 (CID) Co-seismic Ionosphere disturbances

The Tec data shows some variation due to the motion of satellites. This can also happen owing to diurnal variations in the solar zenith angle and long-term disruptions. LTSID, for instance, are large-scale traveling ionosphere disturbances. For the elimination of long-term variation, high pass filters are used. To study the pre-seismic anomalies, two approaches were followed. The short-term anomaly and the long-term anomaly. The short-term precursor is 4-5 days of TEC changes before the main earthquake. Moreover, it can be found in the 2008 Wenchuan earthquake. The long-term anomaly observes 15 days before the main shock. The median of 15 days is calculated. If the measured TEC is larger or less than UB and LB during a calm geomagnetic period, the anomaly is regarded as noteworthy.

2.6 (SIP) sub-Ionosphere point and (IPP) Ionosphere pierce point

The ionosphere is thought to be a thin layer at a set height from the ground, which is often supposed to be 300 kilometers. The junction of the satellite's line of sight and this Ionosphere layer is known as the Ionosphere pierce point (IPP). Furthermore, the projection of this (IPP) on the earth's surface considers a Sub-Ionosphere point (SIP). When examining the geographical aspects of the TEC anomalies, the SIP trajectory is crucial.

When the satellite height is near the horizon, the SIPs are roughly 1000 km distant from the terrestrial GPS receivers. The penetration angle of the line of sight vector to the hypothetical layer of the ionosphere can be calculated. These angles can be used in converting to vertical TEC from slant TEC.

2.7 GIM (Global Ionosphere Map)

The Global Ionosphere Map (GIM), is a scattered map of vertical TEC values at grid points on a global scale. These GIM files are mainly obtained from GPS and GLONASS data. The data comes from International GNSS stations (IGS). These GIM files are also found at many analysis centers such as The University of Berne in Switzerland which is home to the Center for Orbit Determination in Europe (CODE). The following figure is a sample of GIM.

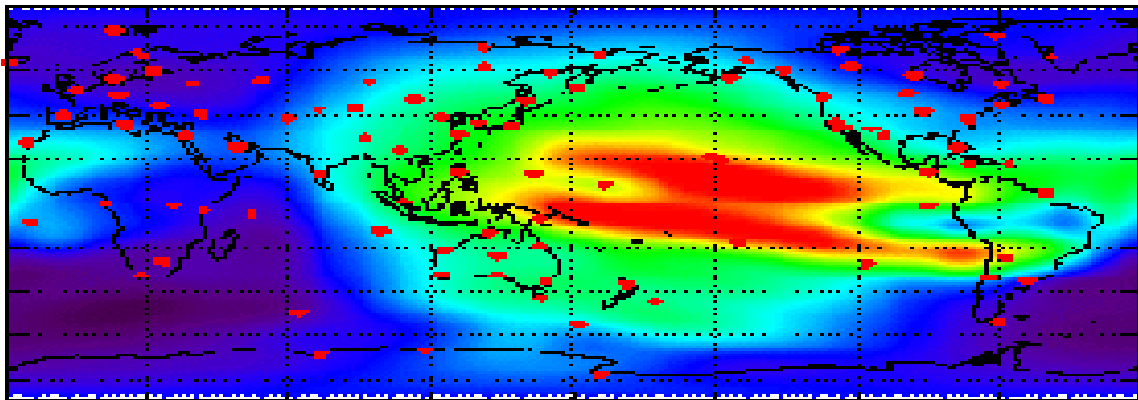


Fig. 2.3 Global Ionosphere Map Example.

(Courtesy: arizonaenergy.org)

2.8 Earthquake Phenomenon

Earthquake is a hazardous phenomenon. In a general sense, it occurs due to the abrupt movement of rocks, along the fault lines to release the stress. Moreover, the movements of the earth's tectonic plates built that up. This results in the generation of seismic waves, which propagate through long distances. Earthquakes are classified into three categories. (i) normal faulting (ii) reverse faulting (iii) strikes-slip faulting. Normal faulting occurs when the side of the overriding rock goes down relative to the other side, this happens due to tensile crustal stress. In reverse faulting, the compressional crustal stress causes the

overriding side to push up relative to the other side. In strike-slip faulting, the crustal stress has tensile in one direction and compression in a perpendicular direction, as result, the rock moves horizontally in the path of the fault strike.

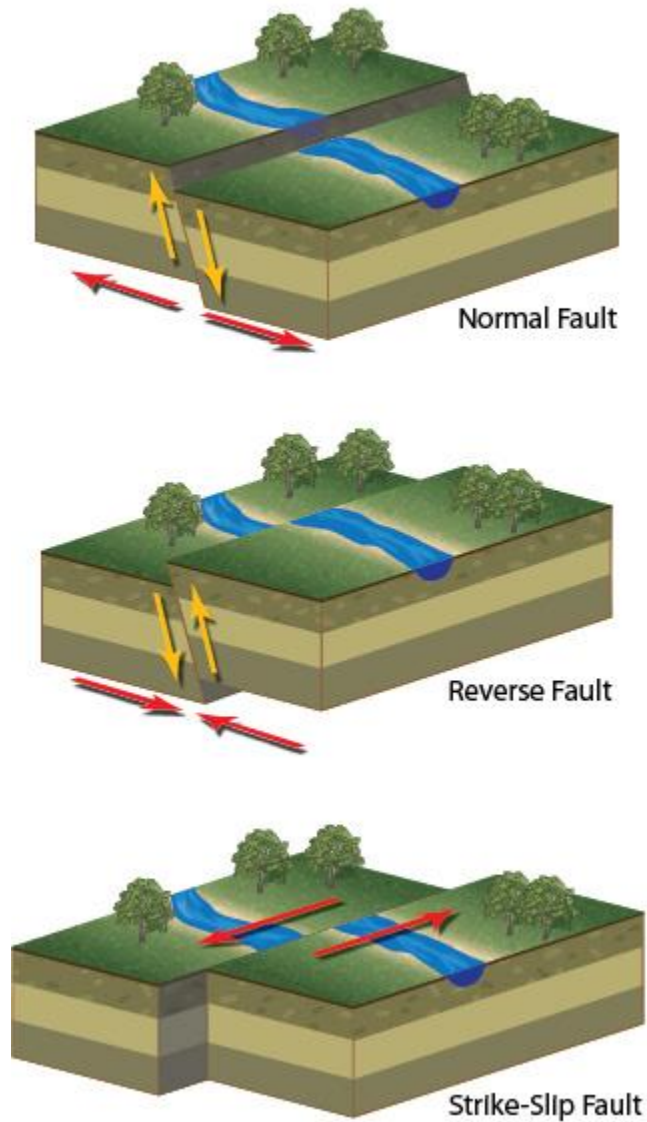


Fig. 2.4 Crustal movement during Earthquake and its types

(Courtesy: www.weather.gov).

2.9 Case Study-2:

The second case investigates Ionosphere and Atmosphere during the period of 11 EQs. Which are greater than magnitude 5. These EQs occurred in 2018 around the Globe. During 2018 around 2000 EQs occurred and 11 of them were filtered out on two conditions. The primary condition limits the selection to a Quiet geomagnetic period. Which can be described as $K_p \leq 3$, $Dst \leq -20$. IT is due to avoiding the complexity of geomagnetic born anomalies with seismic activity. Many of the EQs had abrupt geomagnetic activity. So they were filtered. The secondary condition requires three International GNSS stations in the EQ preparation zone. This zone can be found by the following Dobrovolsky equation [25]:

$$R = 10^{0.43M} \text{ km} \quad (2.8)$$

After both conditions were applied to 2000 EQs, 11 of them fulfilled the desired criteria. The data of all EQs were taken from USGS (United states geological survey). Furthermore, GNSS-TEC measurements were obtained from those stations of the IGS network, which operate within the seism genic zone. Fig. 3.11 represents the geographical locations of the selected EQs and their corresponding IGS stations.

3 Results

3.1 GPS TEC anomalies associated with $M_w > 7$ earthquakes in Japan

3.2 Introduction

There are four earthquakes have been chosen for the detection of Tec anomalies during the seismic period. Three are before the main shock and one is after it. The main occurred on 11 march 2011 time, which is not included in this study. The low-intensity earthquakes are between magnitude 7-8 to study the relation between big earthquakes and the small ones. The first occurred 120 km southeast of Funato, japan at 14:32:43 UTC (Universal Coordinated Time) with a magnitude of 7.3 and having 32.0 km depth, on 9 March 2011. The epicenter was located at (38.435°N 142.842°E) latitude and longitude respectively. This earthquake was followed by more EQs close in time. The second was 47 km east of Arai, japan at time 06:15:40 UTC with a magnitude of 7.9 and having a depth of 42.6 km on 11 March 2011. the epicenter was located at (36.281°N 141.111°E). The third one is 272 km ES of Kamaishi, japan at 06:25:50 UTC with a magnitude of 7.7 and having 18.6 km depth. On 11 March 2011. The epic center was located at (38.058°N 144.590°E). The fourth earthquake is 29 km ES of Ishinomaki, japan at 14:32:43 UTC with a magnitude of 7.1 and having a depth of 42.0 km, on 7 April 2011. The epicenter was located at (38.276°N 141.588°E). The TEC measurement data were taken from three International GNSS stations, which can be used for finding Seismo ionosphere anomalies of these EQs, the stations are close to the epicenter of all four earthquakes.

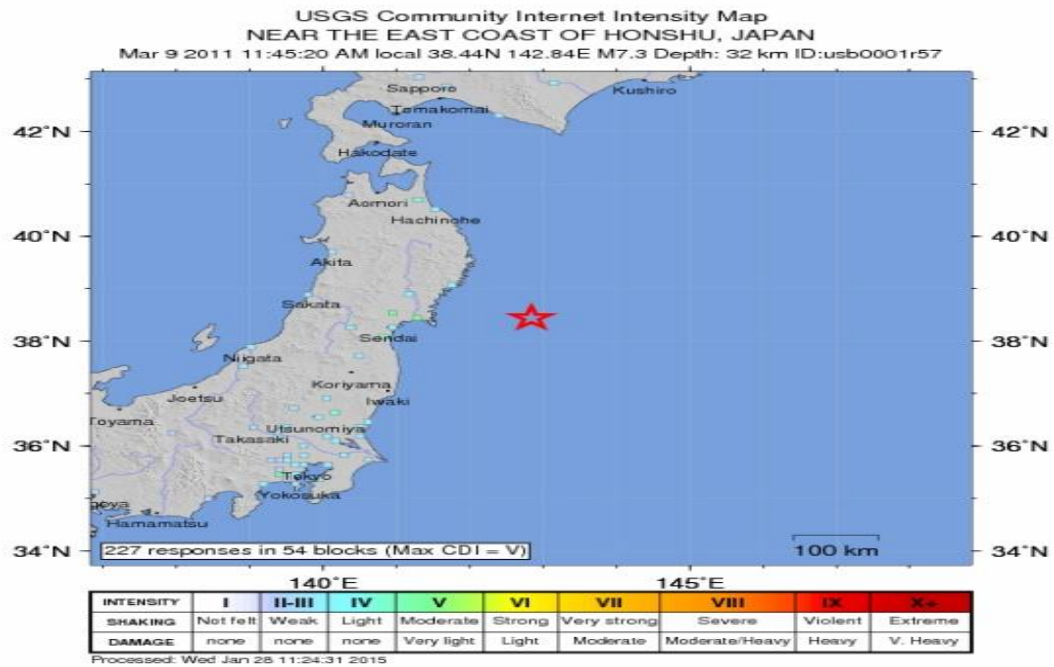


Fig.3.1 Epicentral point of 120 km southeast of Funato, japan, at local 38.44N, 142.84E.
which occurred on 9 March 2011.

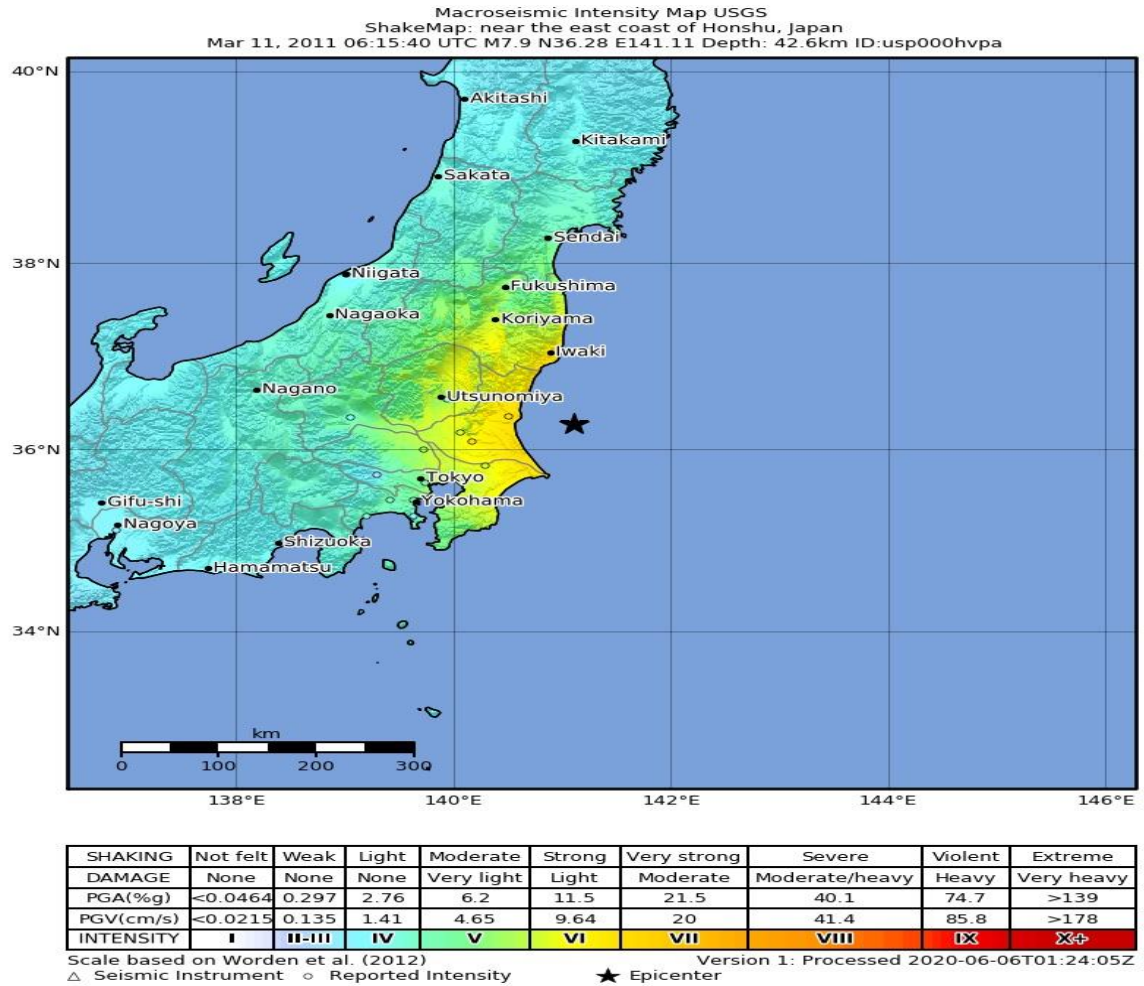


Fig. 3.2 Epicentral point of 47 km east of Arai, Japan, which occurred at 36.28N, 141.11E. Time 06:15:40 UTC on 11 March 2011.

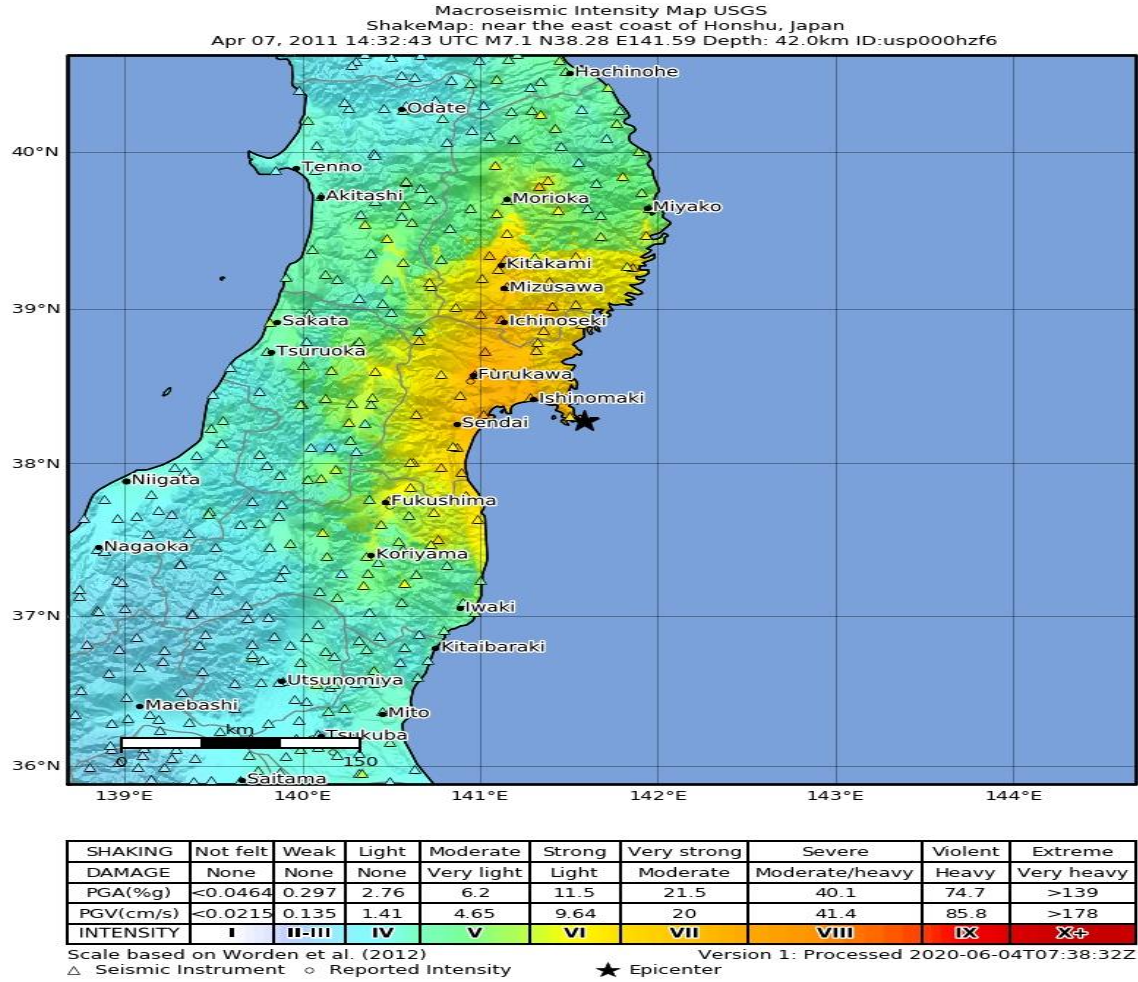


Fig (3.3) Epicentral point of 29 km ES of Ishinomaki, japan at 38.28N, 141.59E.
Occurred on 7 April 2011.

3.3 TEC changes with depth and Magnitude

We have investigated the ionosphere response to the sequential EQs in Japan region that can be regarded as possible seismic precursors. The individual graphs for each analyzed IGS station along with the seismic events are represented as AIRA in Fig. (3.4), KSMV in Fig (3.5), and MATKA in Fig. (3.6). In each of these analyses, we can observe a characteristic ascendancy peak preceding each seismic event that attains its maximum phase within 5 days before the impending shocks but it is associated with the geomagnetic storm, as Kp and Dst remain active. This specific variation observes in each analyzed station in Japan. To get a further zoomed look, we designate the daily TEC values for these stations that are represented in Fig. (3.7). One can see that the TEC values began to climb

in the first two weeks of March 2011. The TEC values gradually ascend from March 1 and reach the maximum until the two gigantic seismic events on March 15, 2011, and subsequently turn back to normal. This unique feature is described for seismic events in the reports [10] and [9]. These variations are associated with the geomagnetic storm. Similarly, the trend was also sounded to be repeating in the case of the April 11 event, where the TEC values again ascend characteristically from April 1 and attain their maximum peak on the main shock day before turning back to normal. As there is a geomagnetic storm these days of Seismic activity, the relation of magnitude and depth cannot be associated clearly with TEC changes. In this sense, we can say that the observed variations are for somehow decoding a trend of seismically induced TEC variations that can open the doors to earthquake forecasting.

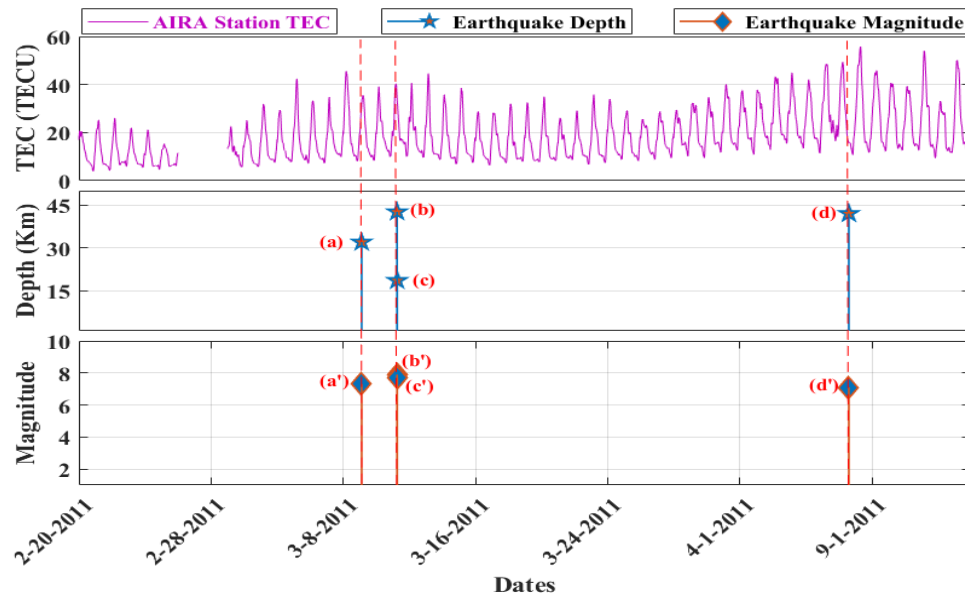


Fig (3.4) Aira station observations with depth and magnitude along with TEC analysis. The TEC quantity started to arise from March 1 and gradually increases until a spike just a day before 9 march EQ but with a depth of 32 km. The next day the spike comes a little lower with consistency throughout the EQ days. On 3 April TEC quantity gradually up until the spike one day after 7 April EQ.

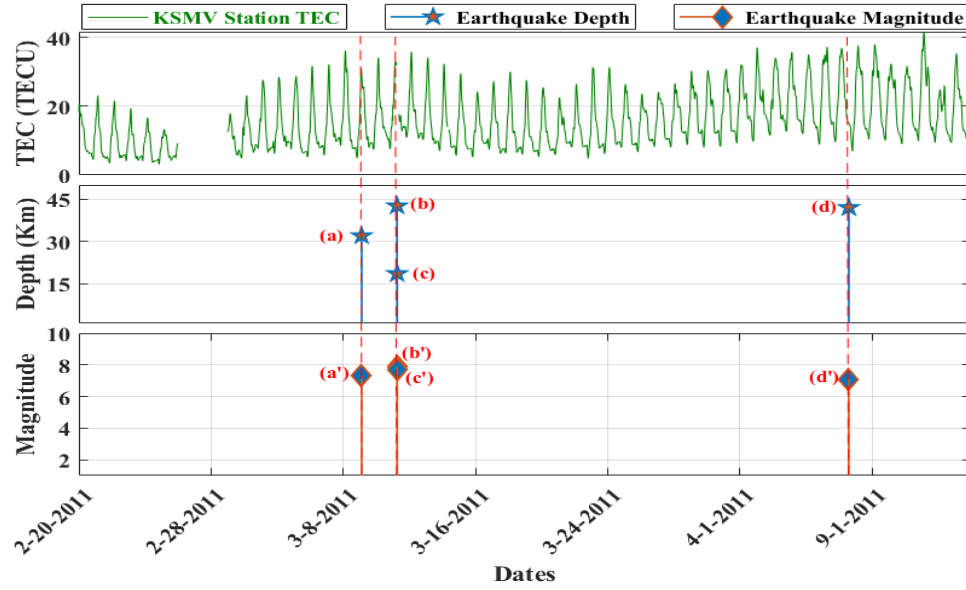


Fig (3.5) KSMV station observations with depth and magnitude along with TEC analysis. The TEC quantity started to arise from March 1 and gradually increases until a spike just a day before 9 march EQ but with a depth of 32 km. The next day the spike comes a little lower with consistency throughout the EQ days. On 2 April TEC quantity a sudden spike and maintain the momentum up to days after the 7 April EQ.

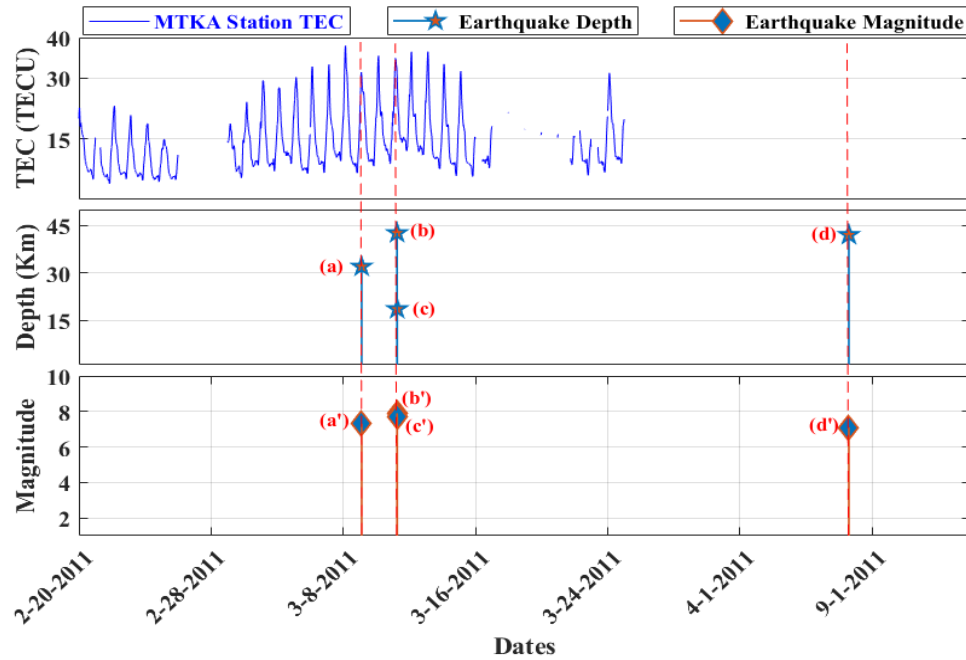


Fig (3.6) MTKA station observations with depth and magnitude along with TEC analysis.

Here from 3 march a gradual increase in TEC quantity and a big spike just a day before 9 march EQ. This abnormal activity of TEC comes down to normal just 4 days of 11 march EQs. The April data here is missing from the MTKA Station.

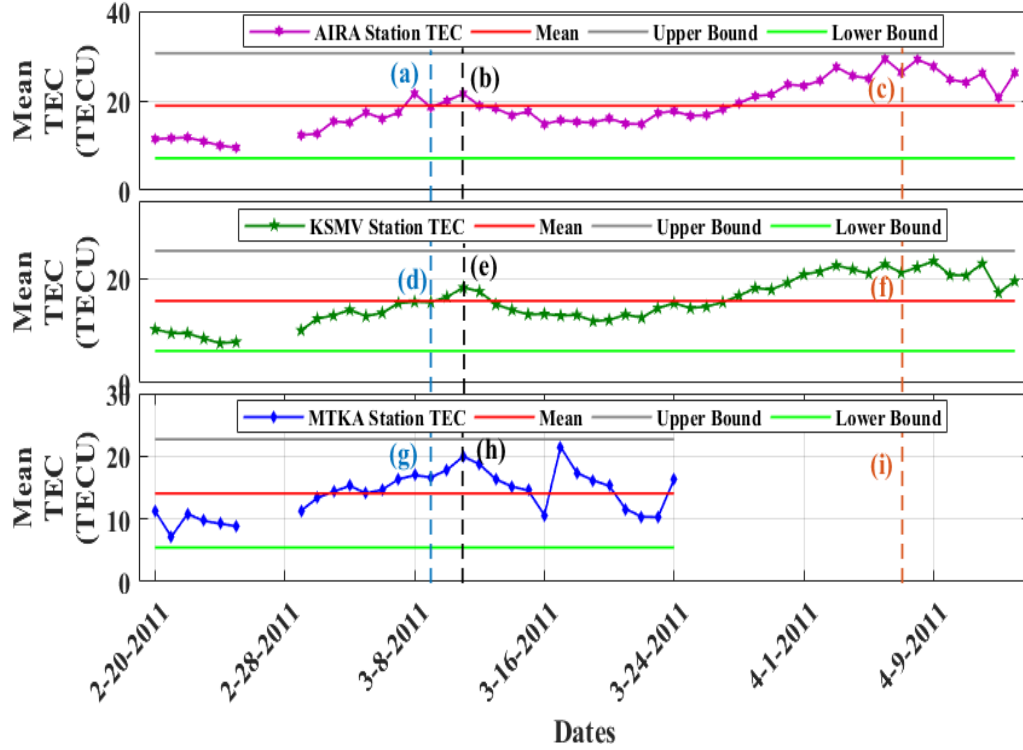


Fig. 3.7 Mean TEC from all stations with upper bound and lower bound. The AIRA and KSMV Stations show somehow concurrent TEC values during the seismic period of March and April 2011. While the MTKA station shows much above from mean TEC value during March Seismic period.

3.4 Co-Seismic Ionosphere disturbances (CID)

The cumulative TEC graphs for all the measured values in the selected IGS stations are shown in Fig (3.8), where the abrupt variations before and after the main shocks are statistically observed along with the series of EQs in March and April associated with the main shock. At KSMV Station, the recorded TEC value started to increase on 6 March at 40 TECU. Moreover, continues gradually until the day of the 3rd EQ which has a magnitude (). then suddenly drop to a normal level. While in April, the TEC value started at a value of less than 40 TECU. Then continues to increase until 8 April, a day after the EQ of 7 April 2011. At AIRA Station, the Recorded TEC value started abruptly at a value of 60

TECU on 7 March. However, decreases gradually at a steady pace up to 12 March. A day after the third EQ occurred on 11 March 2011. The MTKA station recorded the March TEC values in 2011. Which shows the abruption in TEC values on 6 March above 40 TECU and have a normal on 8 March but then decreases with higher TEC value until 11 March 2011.

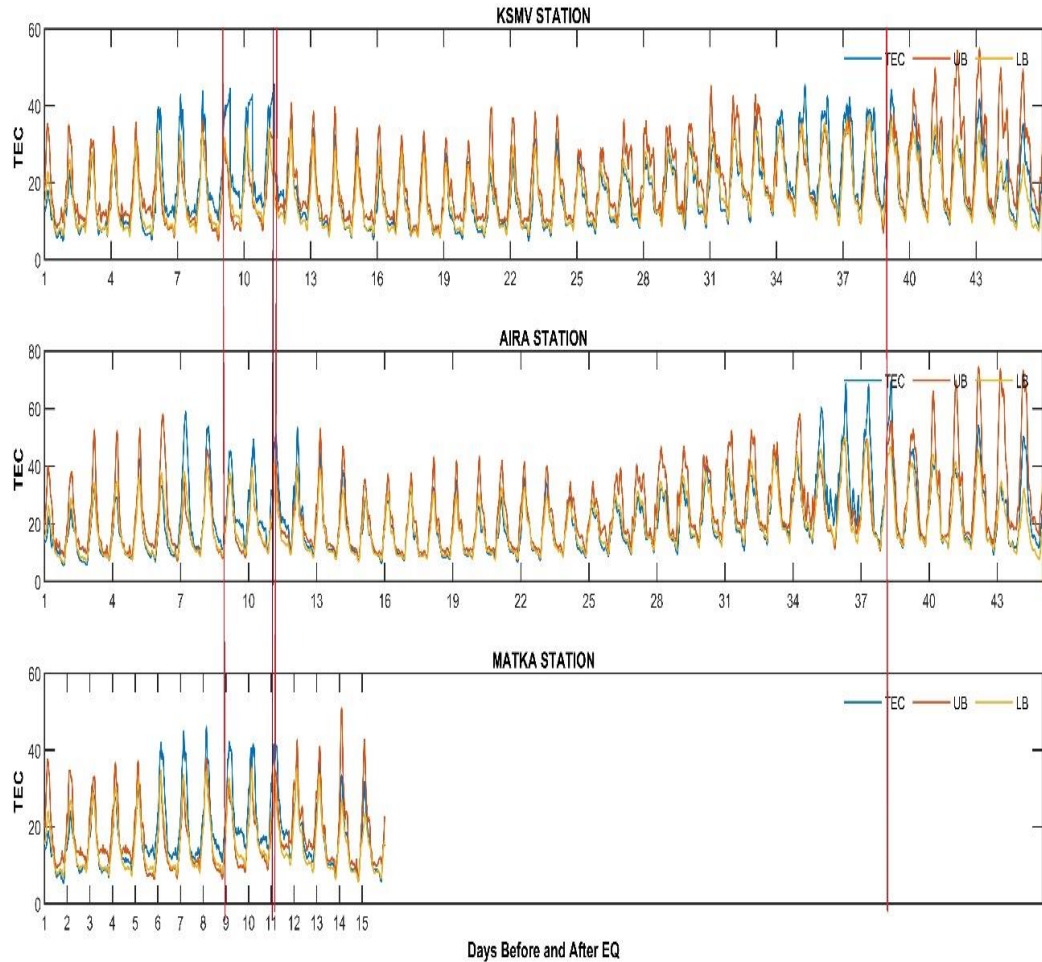


Fig. 3.8 The Analysis of Total Electron content from March 1 to 14 April 2011. For the Stations of AIRA, KSMV and MTKA, Japan.

3.5 Geomagnetic indices

The geomagnetic conditions are closely monitored during the period and the periodicity in Dst and Kp is not perfectly correlating with the TEC values (Fig. 3.9), as we can observe in the case of seismic events to TEC values. The Dst and Kp are not quiet these days. The first EQ occur on 9 march with a Dst less than the threshold for a Geomagnetic storm and with a Kp index less than 2. The consecutive two EQs, which occurred on 11 march have a geomagnetic storm period. The EQ on 7 April also values geomagnetic activity, which affects the clarity regarding TEC association with the occurrence of EQ.

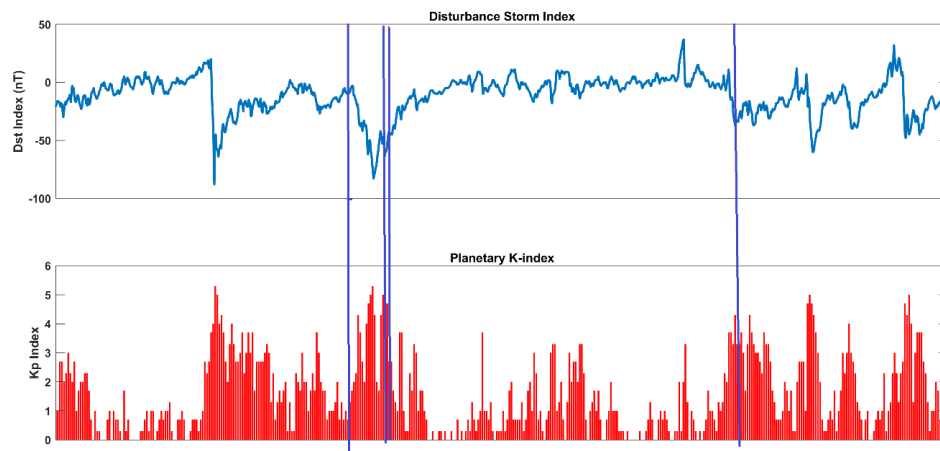


Fig. 3.9 Kp and Dst index during the observed Period from Feb 20 to 14 April 2011.

3.6 TEC Drift

On the day of the big main shock, the Total Electron content was drifting south from Japan. The map shows the TEC concentration in the southern part of Japan while the EQ occurs to the north of the concentration of TEC. The Ionization in the ionosphere starts around 4:00 am and continues to the occurrence of the EQ around 6:00 am.

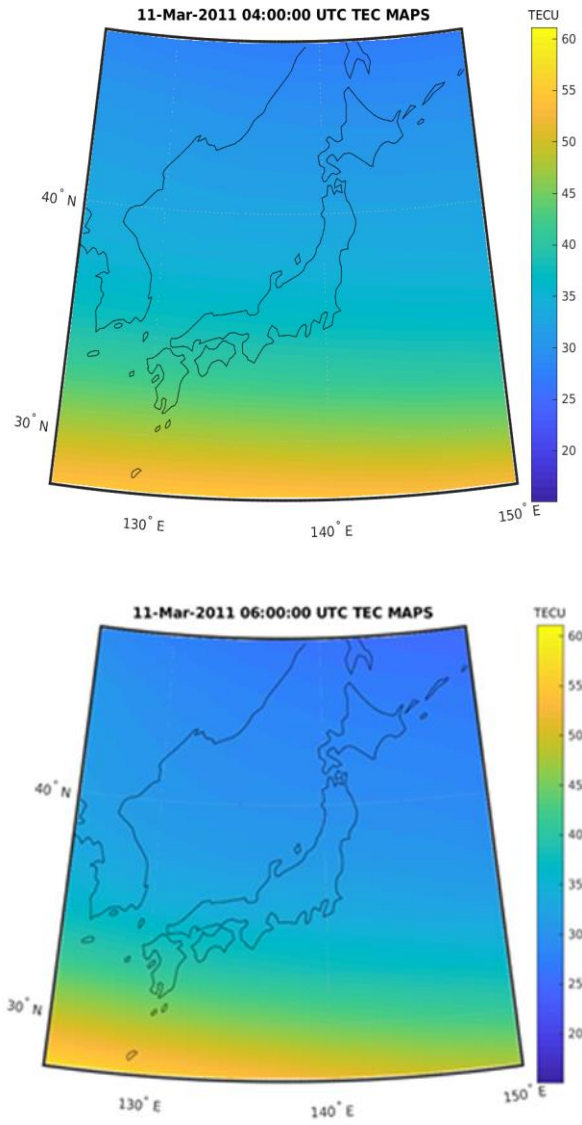


Fig. 3.10 Total Electron Content drift over Japan during the seismic period.

3.7 Possible Seismo-Ionospheric Anomalies Associated with the Earthquakes occurred in 2018 using GNSS TEC: A Statistical Analysis

3.8 Introduction

In this chapter, we study the Ionospheric variations from TEC data and the behavior associated with the earthquake. There are 11 EQs chosen from the year 2018 of different regions around the globe. There were many EQs but 11 stand out on two specific conditions. The first condition is the availability of a minimum of three IGS Stations near the epicenters in the preparation zone for the earthquake. Moreover, the second condition limits that the EQ must be in the storm-less geomagnetic period. This condition is described as that $K_p \leq 3$ and $Dst \geq -20$. The following are the 11 EQs around the Globe. Which occurred in the year 2018. The EQs occur in different regions, which include Alaska, Argentina, Chile, Greece, Russia, Indonesia, and Taiwan. These are the six regions, which are used for assessing the relation of Seismic Ionospheric anomalies. The differential TEC value is analyzed along with diurnal TEC for finding the real cause of TEC anomalies for the two Seismic reasons, whether it is due to seismic events or the geomagnetic Storm effect.

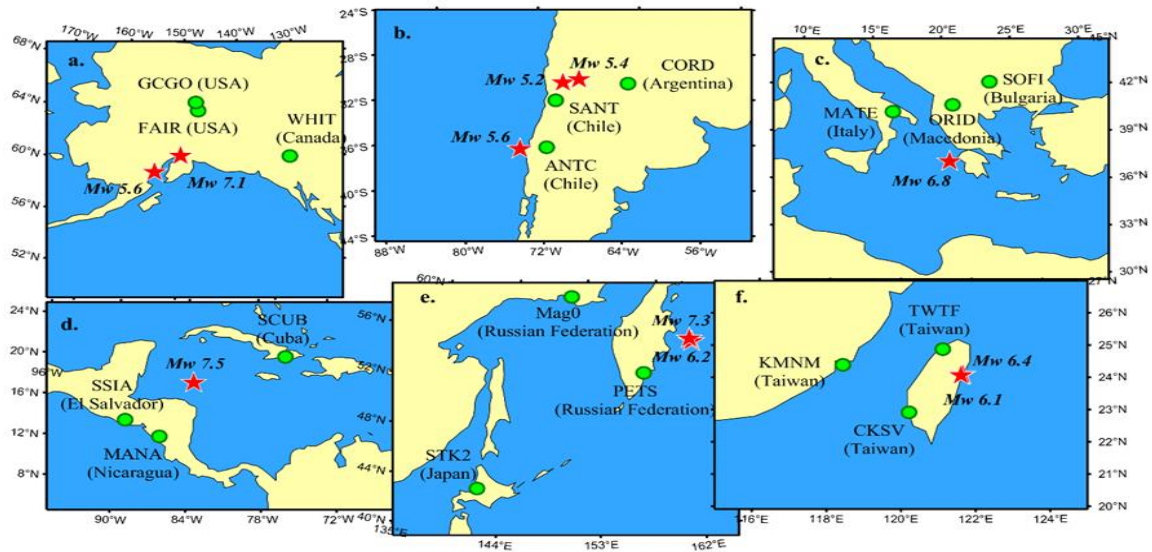


Fig. 3.11 Total Electron Content over Japan during the Seismic Period.

3.9 Siesmo Ionospheric anomalies

In the Alaska region, two EQs occur where a magnitude of 7.1 around 30 Nov 2018, about 14 km Northwest of Anchorage Alaska. The depth is 47 km. located at 61.346N and 149.955 W in latitude and longitude. Just 9 days before another EQ a shock of M 5.6 occurred at 143.3 km depth on 21 Nov 2018. The geomagnetic activity during this period is shown in fig. where the Kp index surpasses the threshold of 3 value on 3 December 2018. An after 7.1 magnitudes of EQ, showing the geomagnetic activity. In the second case, the ΔTEC values are 1.9661, 1.4762, and 1.4103, a day before for the station FAIR, WHIT, and GCGO separately. In the Case of Mw 7.1, ΔTEC values are 1.5995, 1.516, and 1.2682 with 9 % TEC deviation of 36.71 %, 26.7 %, and 73.20% were observed on three IGS stations respectively just three days before.

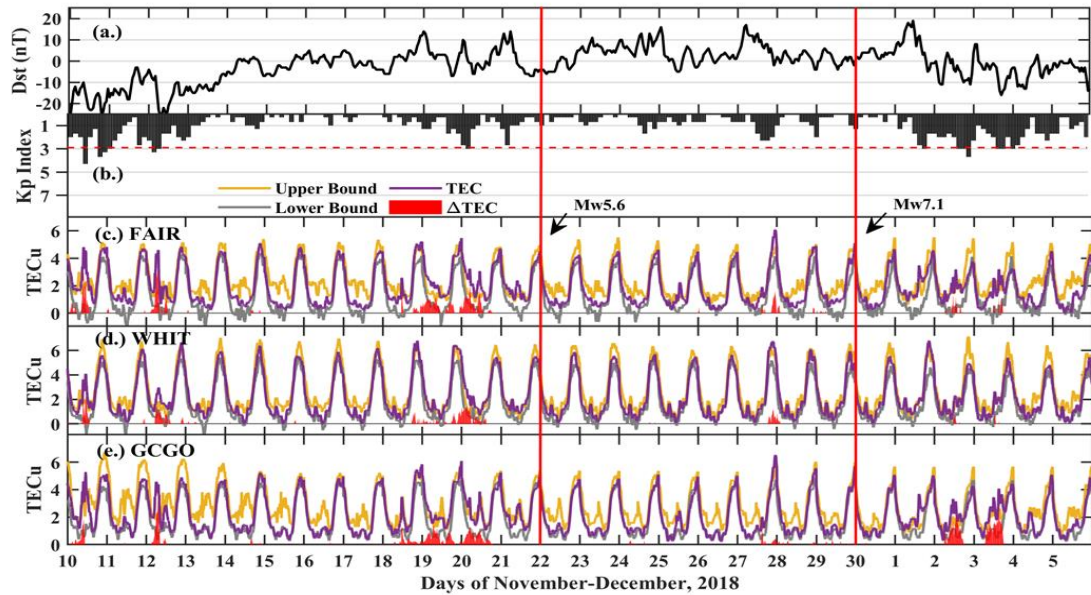


Fig. 3.12 GNSS TEC from stations (FAIR, WHIT, GCGO) for Alaskan EQs of 2018, along with geomagnetic activity in terms of Kp and Dst Index.

The ΔTEC and percentage TEC values of the EQs that occurred in 2018 are given in the table. The geomagnetic activity is at a different time from the main shock period.

Table 3.1: Overall percentage Δ TEC deviations for seismic ($K_p \leq 3$) and storm ($K_p > 3$) anomalies

	Event	Date	Ave. Max. ΔTEC of SIA	Ave. Max. %TEC for SIA	Geomagnetic ally Disturbed Day	Ave. Max. ΔTEC for GIA*	Max. Ave. %TEC for GIA	Value of Kp Index
1	5.6	2018-11-21	1.7193	59.62 %	2018-11-12	2.7431	88.35 %	4-
2	7.1	2018-11-30	1.3756	52.20 %	2018-12-03	2.0459	69.35 %	4-
3	5.4	2018-01-28	6.5606	33.54 %	2018-01-24	5.6546	26.68 %	4o
4	5.2	2018 - 02-01	5.4180	36.41 %	2018-01-24	5.6546	26.68 %	4o
5	5.6	2018-02-08	6.1690	33.13 %	2018-01-24	5.6546	26.68 %	4o
6	6.8	2018-10-25	3.3180	26.58 %	2018-10-13	N/A	N/A	4+
7	7.5	2018-01-10	7.5472	54.41 %	2018-01-08	4.3924	38.96 %	4-
8	7.3	2018-12-20	1.9925	39.05 %	2018-12-09	1.1133	29.18 %	4-
9	6.2	2018-12-25	1.7730	41.45 %	2018-12-09	1.1133	29.18 %	4-
10	6.1	2018-02-04	6.0543	42.75 %	2018-01-24	2.7532	19.82 %	4o
11	6.4	2018-02-06	9.0018	45.69 %	2018-01-24	2.7532	19.82 %	4o

3.10 Atmospheric effect

Some researchers show that air ionization occurs some days before EQ in its surrounding atmosphere Pulinets [30]. During the event of EQ, the crustal rocks collies, which results in the emission of radon gas into the atmosphere. Which accounts for the air ionization phenomenon. The process of air ionization results in the decrease of relative humidity and enhances the thermal budget of the atmosphere in the form of OLR. To confirm the development, anomaly maps of relative humidity and an OLR of the National Oceanic and Atmospheric Administration were examined. Fig () shows the 20% decrease in humidity and the OLR anomalies increased to 30 W/m² in the surrounding epicenters. This gives us the confirmation of air ionization over the epicenter.

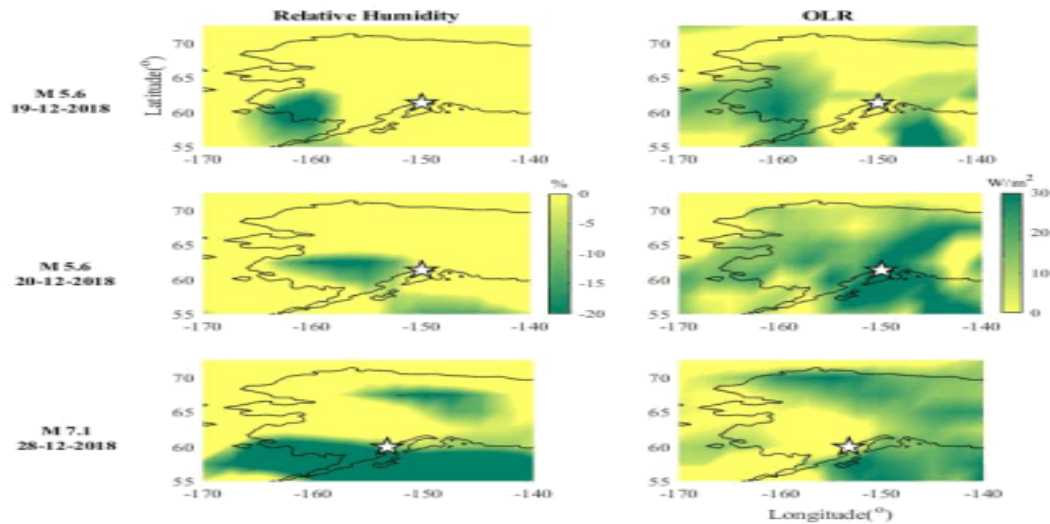


Fig. 3.13 the anomaly maps of humidity and OLR of air ionization in the vicinity of Epicenters of the Alaskan EQs occurred on 19, 20, and 28 December 2018. The white stars show the epicenters.

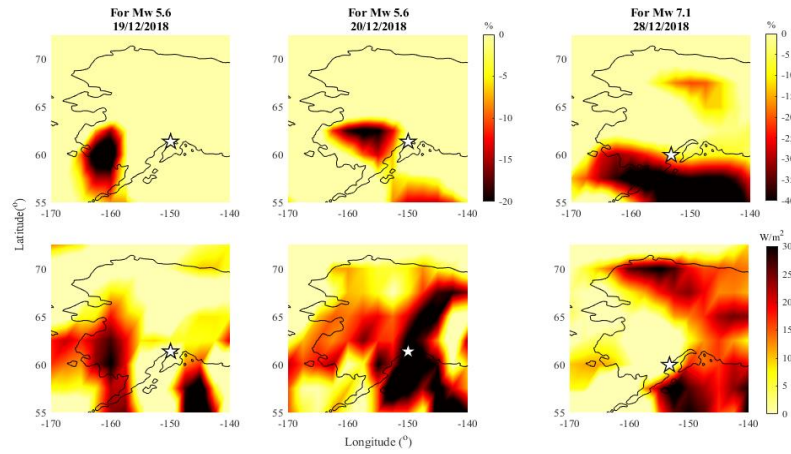


Fig. 3.14 Humidity and OLR of the Alaskan EQs over the Epicenters.

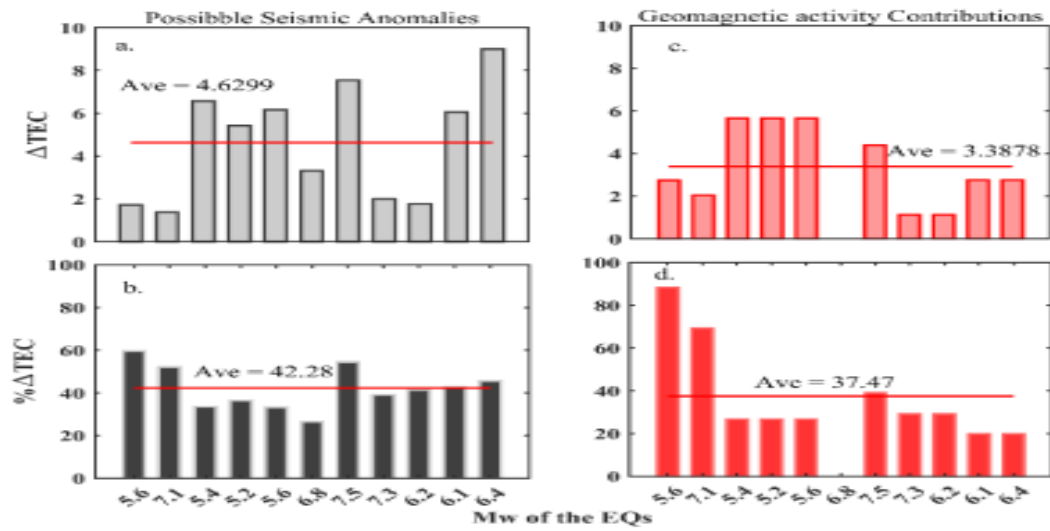


Fig. 3.15 a) shows the possible Seismic anomalies. b) shows the percentage of differential TEC of all EQs. c) shows the geomagnetic activity contribution. d) shows the percentage abruption in all EQs.

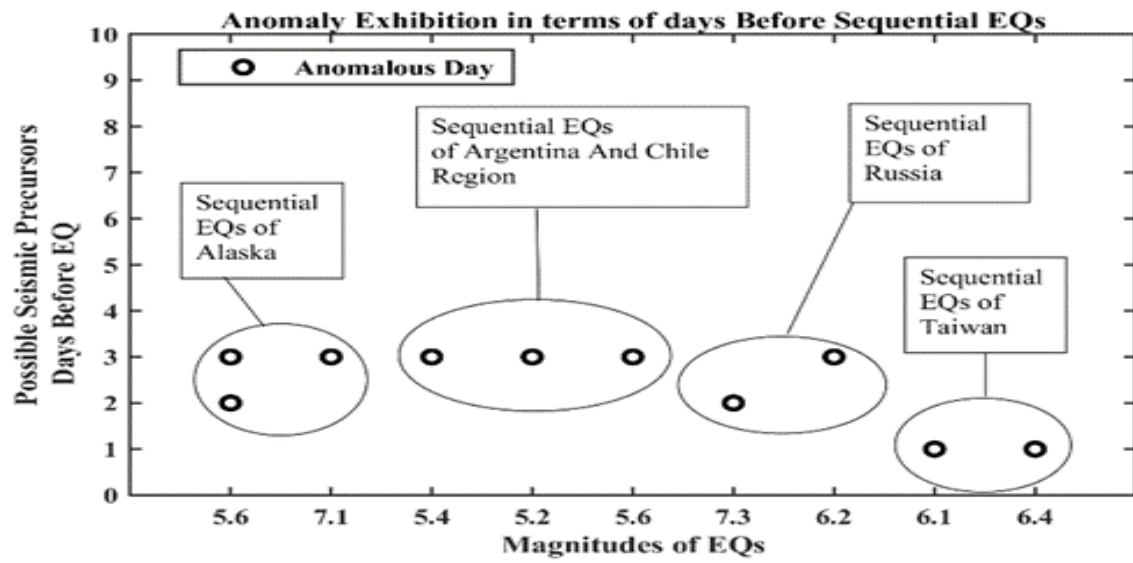


Fig (3.16) the day anomaly pattern of all EQs in respective regions.

4 Conclusions and Future Work

The analysis of Ionospheric perturbations during Seismic events, tells a connection between them. In one study, the results suggest that TEC variations during the geomagnetic activity of $KP \leq 3$ are due to air Ionization, which is produced by Radon Gas emission around the epicenter of EQs. Which resulted in the Ionospheric perturbations through the decrease of humidity and increase of thermal budget in the Atmosphere. In another study, the TEC abruption occurs during geomagnetic activity. Which overshadow its results and elude the TEC variations with EQs during March and April 2011 in Japan. The future prediction of time, epicenter, depth, and magnitude of any EQ has some connection with GNSS TEC variations. Which could help up to some extent to predict the crustal movement Event. For reliable forecasting of EQ, more options of ground and space need to be explored to explain the phenomenon.

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